INCH-POUND
MIL-PRF-62412B
24 November 1995
SUPERSEDING
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5 February 1992

# PERFORMANCE SPECIFICATION

# PUMP ASSEMBLY, OIL, STARTING ENGINE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers a positive displacement, power driven, five element rotary pump which is referred to herein as the pump assembly (see 6.1).

## 2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E/BLUE, Warren, MI 48397-5000 by using the Standardization Document Improvement Proposal (DD Form 1426), appearing at the end of this document, or by letter.

AMSC N/A FSC 2990

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2.2 Government documents.

2.2.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

DEPARTMENT OF DEFENSE

MIL-L-7808 - Lubricating Oil, Aircraft Turbine Engine, Synthetic Base, NATO Code Number 0-148.
MIL-L-23699 - Lubricating Oil, Aircraft Turbine Engines, Synthetic Base.

STANDARDS

DEPARTMENT OF DEFENSE

MIL-STD-810 - Environmental Test Methods and Engineering Guidelines.

(Unless otherwise indicated, copies of the above specifications and standards are available from the Standardization Document Order Desk, 700, Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified. the issues are those cited in the solicitation.

DRAWING

DEPARTMENT OF DEFENSE

12286924 - Pump, Assembly, Oil-Main Lube & Scavenge.

(Copies of these drawings are available from the U.S. Army Tank-automotive and Armaments Command, AMSTA-TR-E/BLUE, Warren, MI 48397-5000.)



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2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ASQC/ANSI Z1.4 - Sampling Procedures and Tables for Inspections by Attributes (DoD Adopted).

(Application for copies may be obtained from the American National Standards Institute, 11 West 42nd Street, New York, NY 10036.)

AMERICAN SOCIETY FOR TESTING AND MATERIAL (ASTM)

ASTM B117 - Standard Practice for Operating Salt Spray (Fog) Testing Apparatus.

(Application for copies may be obtained from the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a first article sample shall be subjected to first article inspection (see 4.3).

3.2 Materials. Materials used shall be in accordance with the manufacturer's materials specifications for oil pumps (see 4.6.1). Recovered materials shall be used to the maximum extent practicable.

3.2.1 Metals. All metals used in the construction of the pump assembly shall be of the corrosion-resistant type or shall be suitably protected to resist corrosion when in storage or during normal service life (see 4.6.1). Unless suitably protected against electrolyte corrosion, dissimilar metals shall not be used in intimate contact with each other.



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3.3 Design and construction. The pump assembly shall conform to the manufacturer's design and construction for oil pumps except for the outline dimensions, the interface dimensions and locations and the valve porting which shall conform to Drawing 12286924 (see 4.6.1 and 4.6.2). The individual valves shall be identified in table I.

TABLE I. Valves.

| Valve type | Description |
|---|---|
| Type I | Pressure regulating valve |
| Type II | Check valve |
| Type III | Check valve |
| Type IV | Pressure relief, no. 4 bearing valve |
| Type V | Check valve |
| Type VI | Filter bypass valve |
| Type VII | Programming valve |

3.3.1 Weight. The weight of the pump without valves, fittings, sensor, plugs, and filter shall not exceed 20 pounds [9.6 kilograms (kg)]. The weight of the complete pump assembly shall be equal to or less than 25.35 pounds (11.48 kg) (see 4.6.1 and 4.6.2).

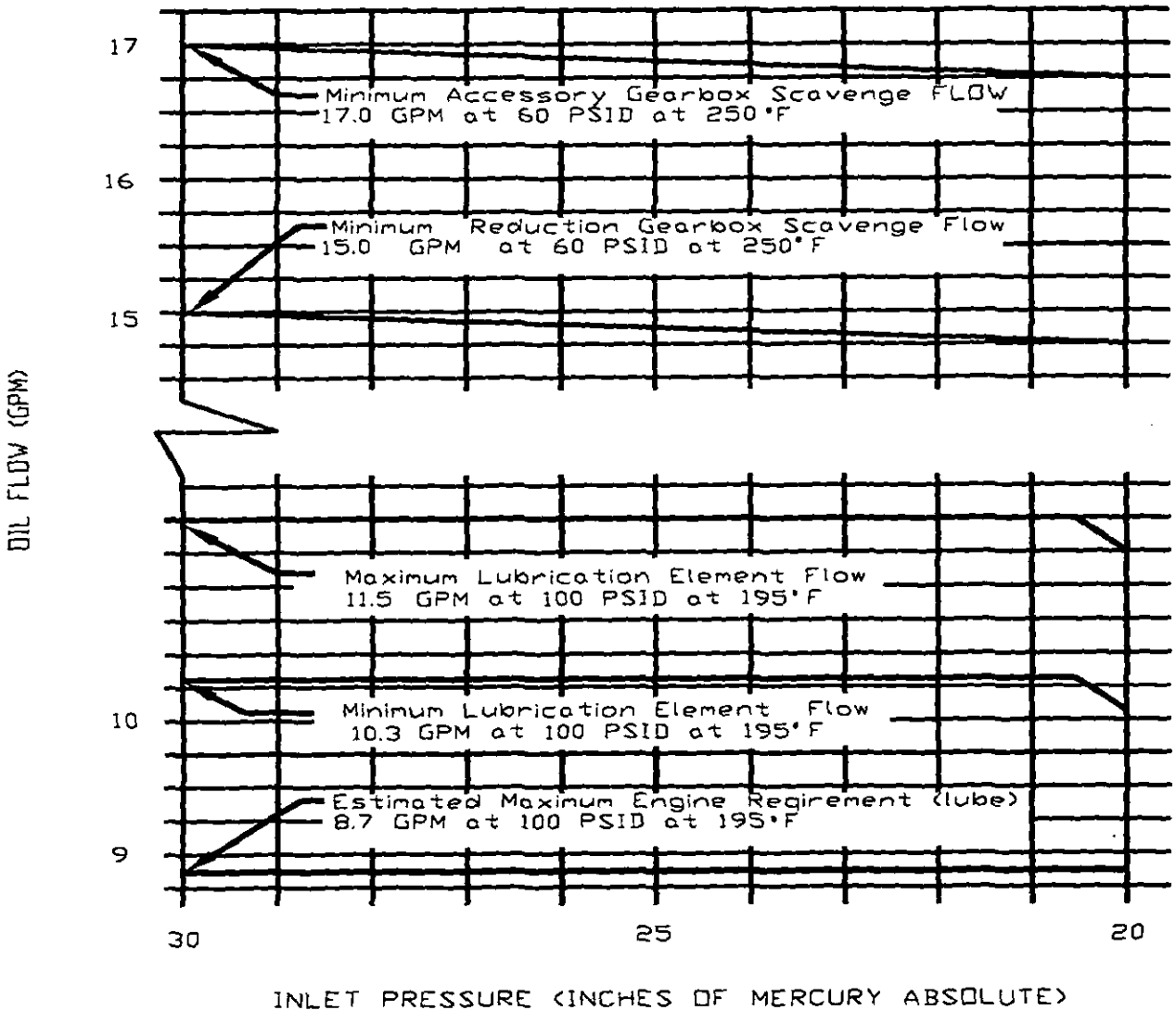
3.4 Performance.

3.4.1 Self-priming. The pump assembly shall be self-priming with dry elements and oil in accordance with 4.6.3 at an inlet pressure of 5 inches (17 kilopascal (kPa)) of mercury (Hg) vacuum and a drive speed of 400 to 600 revolutions per minute rpm). Indications of flow shall be attained within 30 seconds applicable to both pressure and scavenge elements (see 4.6.3).

3.4.2 Oil flows. Scavenge and lubrication element flows shall be in accordance with the requirements of figures 1 and 2 (see 4.6.4).



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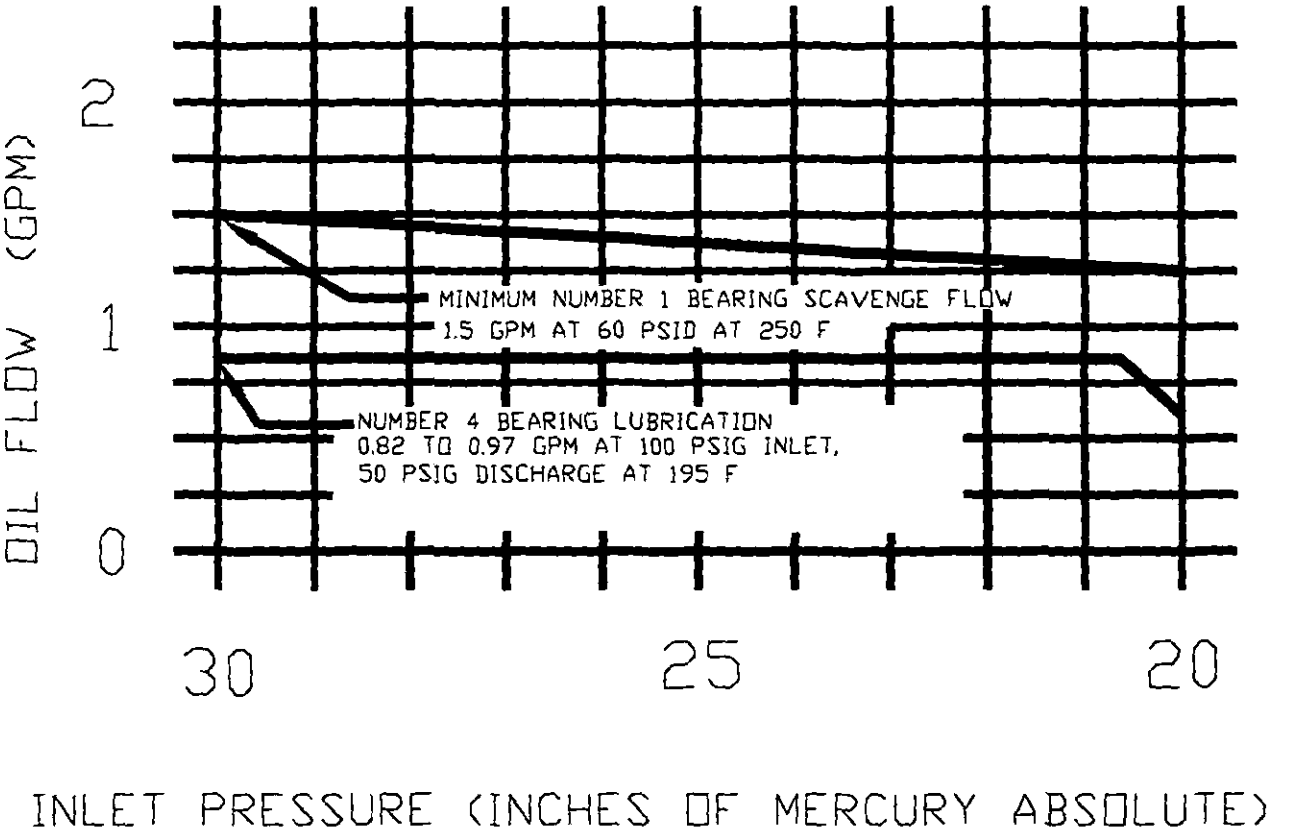

NOTE:

1. The oil may contain up to 10% aeration by volume.
2. The speed shall be 4482 rpm.
3. GPM = Gallons per minute.
4. PSID = Pounds per square inch differential.

FIGURE 1. Gearbox scavenge and lubrication element flows.



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NOTE:

1. The oil may contain up to 10% aeration by volume.
2. The speed shall be 4482 rpm.
3. PSID = Pounds per square inch differential.

FIGURE 2. Bearing scavenge and lubrication element flows.

3.4.3 Pump performance characteristics and conditions. The pump performance characteristics and conditions shall be as specified in table II (see 4.6.5).

3.5 Environmental conditions.

3.5.1 High temperature. The pump assembly shall meet the specified performance requirements when subjected to an ambient air temperature of 250 degrees Fahrenheit (°F) (121 degrees Celsius (°C)) using MIL-L-7808 oil (see 4.6.6.1).



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TABLE II. Performance characteristics and conditions.

| Type of valve | Temperature (Fluid) 1/ | Rated flow and pressure | Pressure | | | | Leakage 2/ |
|---|---|---|---|---|---|---|---|
| | | | Cracking | Full flow | Proof | Burst | |
| I | -65 to 235°F (-54 to 113°C) | 0.6 to 1.3 gpm (2 to 5 L/min) at 120 psig (827 kPa) (276-600 pph) (125 - 272 kgh) | - | 190 to 240 psid (1310 to 1655 kPa) at 4,600 pph (2087 kgh) | 500 psig (3447 kPa) | 750 psig (5171 kPa) | Not to exceed 7 pph (3.2 kgh) at 20 psi (138 kPa) below cracking pressure at 200°F (93°C) |
| II | -65 to 235°F (-54° to 113°C) | 10.0 gpm (38 L/min) minimum at 5 psid (34 kPa) (4,000 pph) (1814 kgh) | 1.5 to 4.0 psig (10.3 to 27.6 kPa) | 6.5 psid (45 kPa) maximum at 4,600 pph (2087 kgh) at 200°F (94°C) | 500 psig (3447 kPa) | 750 psig (5171 kPa) | Not to exceed one-half pound (0.23 kg) of oil under a static head of 2 feet (61 cm) on the pressure inlet neck valve, at an oil and pump temperature of 165°F (74°C) for 72 hours |
| III | -65 to 350°F (-54°to 177°C) | 32 gpm (121 L/min) minimum at 10 psid (69 kPa) (14,720 pph) (6677 kgh) | 1.5 to 4.0 psig (10.3 to 27.6 kPa) | 16 psid (110 kPa) maximum at 14,720 pph) (6677 kgh) at 200°F (94°C) | 300 psig (2068 kPa) | 500 psig (3447 kPa) | Not to exceed one-half pound (0.23 kg) of oil under a static head of 4 feet (122 cm) on the common scavenge outlet check valve, at an oil and pump temperature of 165°F ( 74°C) for 72 hours. |

pph = Pounds per hour
kgh = Kilograms per hour



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TABLE II. Performance characteristics and conditions. (continued)

| Type of valve | Temperature (Fluid) 1/ | Rated flow and pressure | Pressure | | | | Leakage 2/ |
|---|---|---|---|---|---|---|---|
| | | | Cracking | Full flow | Proof | Burst | |
| IV | -65 to 235°F (-54° to 113°C) | 0.92 gpm (3.4 L/min) minimum at 240 to 260 psig (1655 to 1793 kPa) (425 pph) (193 kgh) | 200 to 240 psig (1379 to 1655 kPa) | 240 to 260 psig (1655 to 1793 kPa) at 425 pph (193 kgh) minimum | 500 psig (3447 kPa) | 750 psig (5171 kPa) | Not to exceed 7 pph (3.2 kgh) at 20 psi (138 kPa) below cracking pressure at 200°F (93°C) |
| V | -65 to 235°F (-54 to 113°C) | 10 gpm (38 L/min) minimum at 25 psid (172 kPa) maximum (4600 pph) (2087 kgh) | 15 to 17 psig (103 to 117 kPa) | 25 psid (172 kPa) maximum 4,600 pph (2087 kgh) minimum at 200°F (93°C) | 500 psig (3447 kPa) | 750 psig (5171 kPa) | One pound (0.45 kg) maximum per hour before valve cracks (15 to 17 psid) (103 to 117 kPa). |
| VI | -65 to 235°F (-54 to 113°C) | 10 gpm (38 L/min) minimum at 30 psid (207 kPa) maximum (4,600 pph) (2087 kgh) | 17 to 21 psig (117 to 145 Kpa) | 30 psid (207 kPa) maximum 4,600 pph (2087 kgh) minimum at 200°F) (93°C) | 500 psig (3447 kPa) | 750 psig (5171 kPa) | Zero leakage up to 14 psid (97 kPa) then 10 cc/min up to valve cracking pressure at 200°F (93°C) |



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TABLE II. Performance characteristics and conditions. (continued)

| Type of valve | Temperature (Fluid) 1/ | Rated flow and pressure | Pressure | | | | Leakage 2/ |
|---|---|---|---|---|---|---|---|
| | | | Cracking | Full flow | Proof | Burst | |
| VII | -65 to 235°F (-54 to 113°C) | See figure 3 | - | - | 500 psig (3447 kPa) | 750 psig (5171 kPa) | 50 pph (23 kgh) maximum at 80 psid (552 kPa) or higher. |

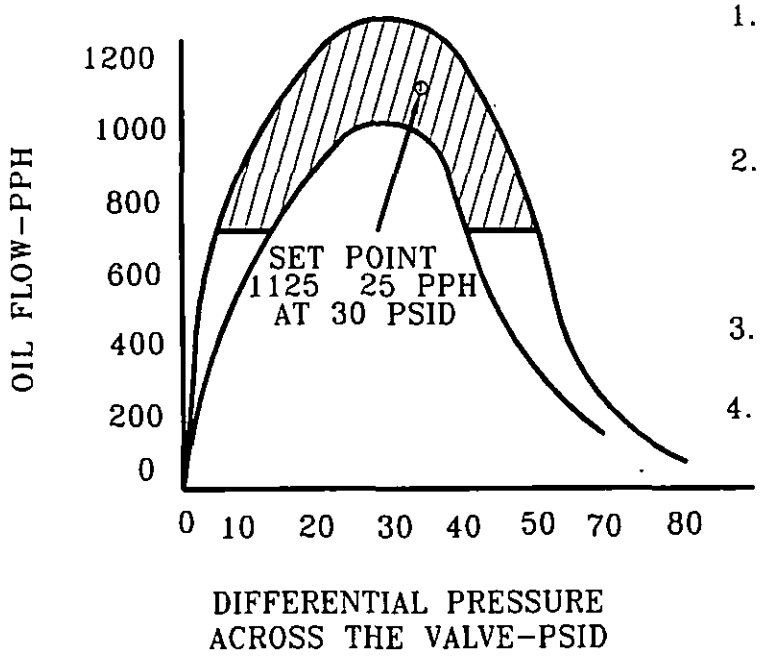
cc/min = Cubic centimeters/minute.

1/ Requirements applicable over the specified temperature range unless otherwise indicated.

2/ Leakage requirement is applicable for the service life of the valve.



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AGT-1500 PROGRAMMING VALUE

DIFFERENTIAL PRESSUSE

VS

BYPASS OIL FLOW

NOTES:

1. BYPASS FLOW SHALL FALL WITHIN SHADED AREA WITH BOTH INCREASING AND DECREASING PRESSURE.
2. A MAXIMUM CHANGE IN OIL FLOW OF 10% SHALL BE ALLOWED IN SUCCESSIVE CYCLES.
3. MAXIMUM OF 50 PPH LEAKAGE AT 80 PSID.
4. OIL SHALL CONFIRM TO MIL-L-23699 AT 195±5° F

FIGURE 3. Programming valve pressure/flow relation.



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3.5.2 Low temperature. The pump assembly shall meet the specified performance requirements when subjected to an ambient temperature of -60°F (-51°C) (see 4.6.6.2).

3.5.3 Gun-firing shock. The pump assembly shall meet the specified performance requirements without malfunction when exposed to shocks of 55 ±5 gravity units (g) for 2.5 ± 0.2 milliseconds (ms) duration in each direction along each of three mutually perpendicular axes (see 4.6.6.3).

3.5.4 Basic shock. The pump assembly shall meet the specified performance requirements without malfunction when exposed to shocks of 20 ± 2 g for 18 ms duration in each direction of three mutually perpendicular axes (see 4.6.6.4).

3.5.5 Ballistic shock. The pump assembly shall meet the specified performance requirements without malfunction when exposed to shocks of 200 ± 2 g for 18 ms duration in each direction along each of three mutually perpendicular axes (see 4.6.6.5).

3.5.6 Vibration. The pump assembly shall meet the specified performance requirements without malfunction when exposed to the peak vibration envelope of figure 4 in each of three mutually perpendicular axes. The vibration shall be imposed at a logarithmic sweep rate of 20 minutes per sweep cycle from 25 to 500 to 24 hertz (Hz) (see 4.6.6.6).

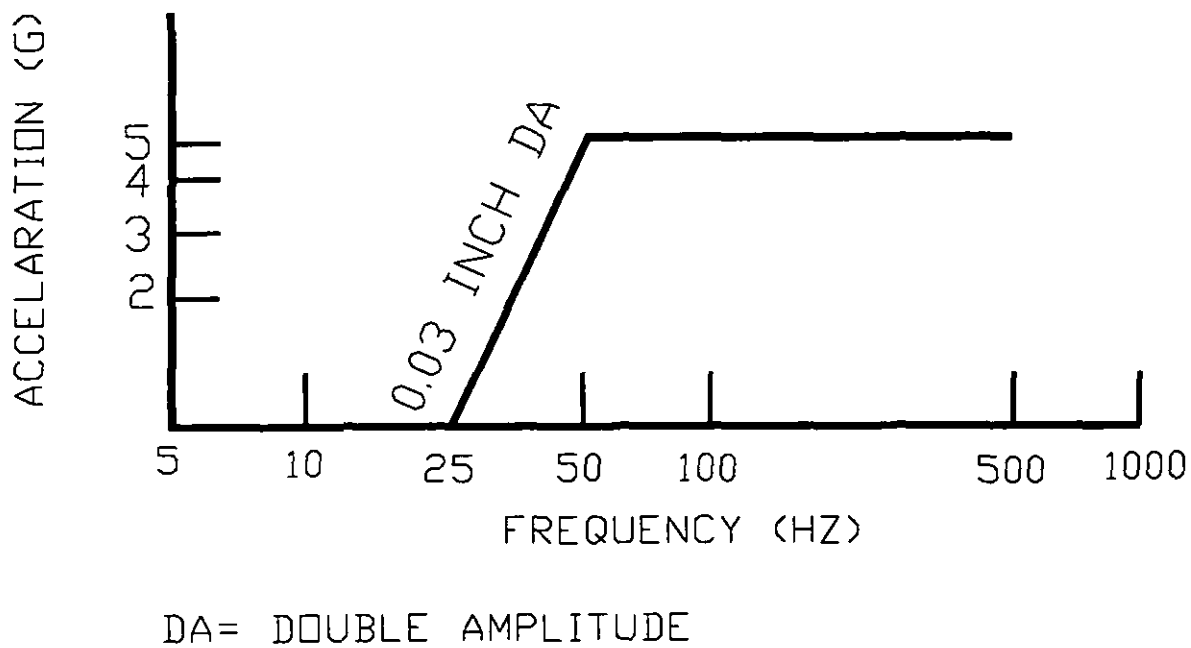



FIGURE 4. Peak vibration envelope.



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3.5.7 Submergence. The pump assembly shall meet the specified performance requirements after submergence in 8 feet (244 cm) of water (see 4.6.6.7).

3.5.8 Sand and dust. The pump assembly shall meet the specified performance requirements after being subjected to a 10 hour sand and dust environment (see 4.6.6.8).

3.5.9 Salt fog. The pump assembly shall meet the specified performance requirements after being subjected to a 48 hour salt fog test (see 4.6.6.9).

3.5.10 Fungus. The pump assembly shall meet the specified performance requirements after being exposed to fungus growth (see 4.6.6.10).

3.6 Identification marking. Identification marking shall include, as minimum, the following information permanently marked on the exterior of the pump assembly (see 4.6.2):

a. Manufacturer's identification
b. Serial number
c. Military part number
d. Federal stock number
e. Manufacturer's part number
f. Date of manufacture
g. Contract number
h. U.S.

3.7 Workmanship. The pump shall be free of defects which may adversely affect the serviceability or appearance of the pump. Particular attention shall be given to the neatness and thoroughness of welding, brazing, marking, plating, painting, and freedom of parts from burrs and sharp edges (see 4.6.2).

4. VERIFICATION

4.1 Classification of inspection.

a. First article inspection (see 4.3).
b. Conformance inspection (see 4.6).

4.2 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be conducted under the following conditions:



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a. Air temperature: 73 ± 18°F (23, ± 10°C).
b. Barometric pressure: 28.5 (+2, -3) inches of Hg.
c. Relative humidity: 50 " 30%.

4.3 First article inspection. Unless otherwise specified (see 6.2), first article inspection shall be performed on two pump assemblies when a first article sample is required (see 3.1). This inspection shall include the examinations of 4.5 (see table III) and the tests of 3.4.1 through 3.5.10 (see table IV).

TABLE III. Classification of defects.

| Category | Defect | Method of examination |
|---|---|---|
| Critical | None | |
| Major: | | |
| 101 | Dimensions affecting interchangeability, not within tolerance (see 3.3). | SIE 1/ |
| 102 | Weight, not as specified (see 3.3.1). | Scale |
| 103 | Identification marking, improper (see 3.6). | Visual |
| 104 | Faulty workmanship affecting performance (see 3.7). | Visual |
| Minor: | | |
| 201 | Dimensions not affecting interchangeability, not within tolerance (see 3.3). | SIE 1/ |
| 202 | Faulty workmanship affecting appearance (see 3.7). | Visual |

1/ SIE = Standard Inspection Equipment.

TABLE IV. Classification of tests.

| Title | Requirement | Inspection | First article | Quality conformance | |
|---|---|---|---|---|---|
| | | | | Examination | Tests |
| Materials, design and construction | 3.2 thru 3.3 | 4.6.1 thru 4.6.2 | X | | |
| Defects (see table III) | 3.3, 3.6, 3.7 | 4.6.1 thru 4.6.2 | X | X | |



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TABLE IV. Classification of tests. (continued)

| Title | Requirement | Inspection | First article | Quality conformance | |
|---|---|---|---|---|---|
| | | | | Examination | Tests |
| Self priming | 3.4.1 | 4.6.3 | X | | X |
| Oil flows | 3.4.2 | 4.6.4 | X | | X |
| Pump performance characteristics and conditions | 3.4.3 | 4.6.5 | X | | X |
| High temperature | 3.5.1 | 4.6.6.1 | X | | |
| Low temperature | 3.5.2 | 4.6.6.2 | X | | |
| Gun-firing shock resistance | 3.5.3 | 4.6.6.3 | X | | |
| Basic shock | 3.5.4 | 4.6.6.4 | X | | |
| Ballistic shock | 3.5.5 | 4.6.6.5 | X | | |
| Vibration | 3.5.6 | 4.6.6.6 | X | | |
| Submergence | 3.5.7 | 4.6.6.7 | X | | |
| Sand and dust | 3.5.8 | 4.6.6.8 | X | | |
| Salt fog | 3.5.9 | 4.6.6.9 | X | | |
| Fungus | 3.5.10 | 4.6.6.10 | X | | |

4.4 Conformance inspection. Conformance inspection shall include the examinations of 4.5 and the tests of 4.6.3, 4.6.4 and 4.6.5.

4.5 Examination.

4.5.1 Sampling. Samples from an inspection lot for conformance inspection shall be selected in accordance with ASQC/ANSI Z1.4. Each sample selected shall be inspected as specified herein for the defects listed in table III. Conformance to a lot shall be accepted when zero (0) defects are found and rejected when one (1) or more defects are found.

4.6 Method of inspection.

4.6.1 Materials, design and construction. Conformance to 3.2 through 3.3.1 shall be determined by inspection of contractor records providing proof or certification that materials conform to requirements. Applicable records shall include drawings, specifications, design data, receiving inspection records, processing and quality control standards, vendor catalogs and certifications, industry standards, test reports and rating data.



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4.6.2 Defects. Conformance to 3.3, 3.3.1, 3.6 and 3.7 shall be determined by examination for the defects listed in table III. Examination shall be visual, tactile, or by measurement with SIE.

4.6.3 Self-priming. To determine conformance to 3.4.1, the pump assembly shall be tested with dry elements and MIL-L-23699 oil at an inlet pressure of 5 inches Hg (17 kPa) vacuum (tank above pump). The pump shall be operated with a drive speed of 400 to 600 rpm while the pump pressure and elements shall be monitored for the length of time it takes for an indication of flow.

4.6.4 Oil flows. To determine conformance to 3.4.2, the pump assembly shall be subjected to a differential pressure versus bypass oil flow test as specified in figures 1 and 2.

4.6.5 Pump performance characteristics and conditions. To determine conformance to 3.4.3, the pump assembly shall be tested under the conditions of table II.

4.6.6 Environmental conditions.

4.6.6.1 High temperatures. To determine conformance to 3.5.1, the pump assembly shall be heat-soaked at -25°F (121°C) per procedure I (storage), method 501 of MIL-STD-810.

4.6.6.2 Low temperature. To determine conformance to 3.5.2, the pump assembly shall be cold-soaked at -60°F (-51°C) per procedure I (storage), method 502 of MIL-STD-810.

4.6.6.3 Gun-firing shock resistance. To determine conformance to 3.5.3, the pump assembly shall be mounted using the installed interface configuration for its intended application or equivalent and tested in accordance with procedure IV (transit drop), method 516 of MIL-STD-810, except in accordance with requirements of 3.5.3. A total of three half-sine wave shock pulses shall be applied in both directions along each of three mutually perpendicular axes. Peak amplitude shall be 55 $\pm$ 5.5 g for 2.5 $\pm$ 0.2 ms. During exposure, the pump shall operate satisfactorily in accordance with the requirements of 3.4.1 and 3.4.2.

4.6.6.4 Basic shock resistance. To determine conformance to 3.5.4, the pump assembly shall be mounted rising the installed interface configuration for its intended application or equivalent and tested in accordance with procedure IV (transit drop), method 516 of MIL-STD-810, except that the conditions of 3.5.4 shall apply. A total of three half-sine wave shock pulses shall be applied in both directions along each of three mutually perpendicular axes. Peak amplitude shall be 20 $\pm$ 2 g for 18 ms. During exposure, the pump assembly shall operate in accordance with requirements of 3.4.1 and 3.4.2.



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4.6.6.5 Ballistic shock resistance. To determine conformance to 3.5.5, the pump assembly shall be tested in accordance with procedure IV (transit drop), method 516 of MIL-STD-810, except that the conditions of 3.5.5 shall apply. The pump assembly shall be mounted in a nonoperational mode and subjected to three half-sine shock impulses of 200 ± 20 g for 0.5 ± 0.1 ms in each direction along each of three mutually perpendicular axes.

4.6.6.6 Vibration resistance. To determine conformance to 3.5.6, the pump assembly shall be mounted using the installed interface configuration for its intended application or equivalent and tested in accordance with procedure I (ground mobile), method 514 of MIL-STD-810, except that conditions of 3.5.6 shall apply. The vibration schedule shall be in accordance with 3.5.6. A 20 minute logarithmic resonant sweep shall be conducted in each of three mutually perpendicular axes, from 25 to 500 to 25 Hz, to identify up to four of the most severe resonant frequencies. A resonant dwell shall then be conducted for 20 minutes at each of the four resonant frequencies. for a total of 100 minutes in each axis. If less than four resonant frequencies are encountered, the balance of the 100 minutes vibration time in each axis shall be spent cycling the pump assembly over the specified frequency range. During exposure, the pump assembly shall operate satisfactorily in accordance with the requirements of 3.4.1 and 3.4.2.

4.6.6.7 Submergence. To determine conformance to 3.5.7, the pump assembly shall be mounted using the installed interference configuration or equivalent, and subjected to a one-hour cycle test consisting of immersion in 70°F (21°C) water to a depth of 6 to 8 feet (183 to 610 centimeters (cm)) at a cycle rate of 4 cycles per minute. Within two hours after completion, the pump assembly shall be tested to verify conformance to the requirements of 3.4.1 and 3.4.2.

4.6.6.8 Sand and dust. To determine conformance to 3.5.8, the pump assembly shall be mounted rising the installed interface configuration, or equivalent, and subjected to a sand and dust environment for a period of 10 hours. A stream of dust-laden air 3.54 to 17.6 gram per cubic meter density shall be directed at the pump at a velocity of 600 to 1200 feet (183 to 366 meters (m)) per minute. Test chamber temperature of 80 ± 10°F (27 ± 3°C) and a relative humidity of 10% maximum, respectively, shall be maintained during the test of 10 hours. The pump assembly shall then be tested to verify conformance to the requirements of 3.4.1 and 3.4.2.

4.6.6.9 Salt fog. To determine conformance to 3.5.9, the pump assembly shall be placed in a test chamber and exposed to a salt fog for a period of 48 hours in accordance with ASTM B117.

4.6.6.10 Fungus. To determine conformance to 3.5.10, the pump assembly shall be inoculated with a spore suspension and tested in accordance with method 508 of MIL-STD-810.



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## 5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

## 6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. The pump assembly is intended to be used as the lubricating oil and scavenge pump on the AGI 1500 gas turbine engine of the M1 tank.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2.1 and 2.3).
c. If first article samples are required (see 3.1).
d. Identification and marking if other than as specified (see 3 .6).
e. Test conditions, if other than as specified (see 4.2).
f. If first article inspection is not required (see 4.3).
g. If initial production inspection is not required (see 4.5).
h. Packaging requirements (see 5.1).

6.3 Supersession. This specification supersedes MIL-P-62412A(AT), dated 5 February 1992.

6.4 Subject term (key word) listing.

Pressure
Revolutions
Scavenge
Self-priming
Valve



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6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - AT
Navy - MC

Preparing Activity:
Army - AT

(Project 2990-0119)



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

## INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

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| I RECOMMEND A CHANGE: | 1. DOCUMENT NUMBER<br>MIL-PRF-62412B(AT) | 2. DOCUMENT DATE (YYMMDD)<br>951124 |
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3. DOCUMENT TITLE
PUMP ASSEMBLY, OIL, STARTING ENGINE

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

| a. NAME (Last, First, Middle Initial) | b. ORGANIZATION | |
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| c. ADDRESS (Include Zip Code) | d. TELEPHONE (Include Area Code)<br>(1) Commercial<br>(2) AUTOVON<br>(If applicable) | 7. DATE SUBMITTED (YYMMDD) |

8. PREPARING ACTIVITY

| a. NAME | b. TELEPHONE (Include Area Code)<br>(1) Commercial<br>(810) 574-8745 | (2) AUTOVON<br>786-8745 |
|---|---|---|
| c. ADDRESS (Include Zip Code)<br>Commander<br>U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E/BLUE,<br>Warren, MI 48397-5000 | IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:<br>Defense Quality and Standardization Office<br>5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466<br>Telephone (703) 756-2340 AUTOVON 289-2340 | |

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